

# Hernan Dario Mojica Diaz



13hernanm@gmail.com | Curriculum Vitae, 22<sup>nd</sup> July 2026

## PROFESSIONAL PROFILE

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Mechanical engineer with experience in applied research, energy systems modeling, and software development for engineering solutions. I have worked on energy transition scenario analysis, climate change mitigation, emissions inventories, and environmental data processing. I have training and experience in data science, artificial intelligence, machine learning, and software development, including architecture design, process automation, AI agents, and systems integration. My profile combines expertise in energy, thermofluids, CFD, and sustainability with the development of technology products.

## EDUCATION

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**Universidad nacional de Colombia**, Bogotá, Colombia

*February 2018 - December 2024*

Bachelor's Degree in *Mechanical Engineering*

Thesis Title: *Fluid dynamic analysis of a miniature propulsion turbine*

Supervisor: Dr. Juan Miguel Mantilla

## SKILLS

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Spoken Languages:

English B2 (Professional)

Programming Languages, Tools, and Technologies:

Python, JavaScript, Git, Linux, Google Cloud Platform (GCP), NestJS, Docker, PyTorch

LLMs and AI Tools:

Deep Learning, Machine Learning, Databases (SQL, NoSQL, and vector databases), MCP, LlamaIndex, Ollama

Engineering Tools and Technologies:

TIMES Model, Autodesk Software, Microsoft Office, Ansys, OpenFOAM

Soft Skills:

Teamwork, Leadership, Problem Solving, Communication, Adaptability

## PROFESSIONAL EXPERIENCE

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**Orion AI**

*March 2026 - Present*

Creator and Technical Lead

- Led the design, development, and evolution of Orion AI from the ground up, a multi-tenant SaaS platform integrating CRM, conversational agents, and WhatsApp sales and support automation.
- Designed a full-stack architecture with Next.js, React, TypeScript, FastAPI, and PostgreSQL, applying asynchronous processing, tenant isolation, and server-first patterns.
- Implemented agents with LangGraph, persistent conversational memory, and tools to query knowledge, update the sales pipeline, and transfer conversations to human advisors.
- Built a RAG system with Qdrant to index PDF, Markdown, and TXT files, with configurable embeddings, per-agent isolated semantic search, and durable workers with retries.
- Integrated Chatwoot, WhatsApp Cloud, and Notion to manage channels, webhooks, voice notes, Agent Bots, and bidirectional lead synchronization.
- Developed the CRM and its operational analytics, including lead tracking, roles and permissions, payment statuses, conversion metrics, and knowledge base management.

- Defined product security, observability, and operations using JWT, RBAC, credential encryption, rate limiting, Prometheus metrics, automated testing, Docker, and deployments on Railway and VPS infrastructure.

## **SatEmis**

*August 2025 - Present*

Software and Artificial Intelligence Engineer - Founder

- Designed and implemented a distributed architecture with NestJS, FastAPI, Docker, and PostgreSQL/PostGIS to manage emissions inventories and geospatial processing.
- Developed a satellite data service with Google Earth Engine and Sentinel-5P to query, aggregate, and analyze methane concentrations by department in Colombia.
- Built APIs for the ingestion, processing, and querying of climate inventories and geospatial data.
- Developed a web application with React and TypeScript integrating authentication, interactive maps, emissions visualization, and internationalization.
- Integrated artificial intelligence capabilities to query and contextualize climate regulations, making environmental information easier to interpret.
- Containerized and optimized services on Google Cloud Platform (GCP) to improve the use of processing and storage resources.

## **Universidad nacional de Colombia**

*February 2024 - July 2025*

Researcher and Energy Systems Analyst

- Developed Colombia's national energy reference model using the VEDA framework for TIMES (IEA-ETSAP).
- Collected and processed data on the Colombian energy system by economic sector and region.
- Processed large datasets and developed analytical tools using Python.
- Projected energy transition scenarios aligned with national climate change plans.
- Led and coordinated multidisciplinary teams.
- Communicated technical results effectively to non-technical audiences.

## **Universidad nacional de Colombia**

*April 2022 - February 2024*

Research Assistant - Faculty of Engineering

- Supported the development of the mitigation section of the Comprehensive Climate Change Plan (PIGCC) for power-sector companies participating in the Carbon-Neutral Electric Sector Alliance (ASECN).
- Quantified Scope 1, 2, and 3 carbon footprints using the GHG Protocol methodology.
- Developed statistical models to project emissions based on operational variables.
- Evaluated mitigation measures such as energy efficiency, fuel substitution, energy mix changes, and carbon offsetting.
- Conducted statistical studies on the relationship between climate phenomena and energy generation by source.
- Prepared presentations for stakeholder engagement and the dissemination of results.

## **TEACHING AND VOLUNTEERING**

### **Universidad nacional de Colombia**

Instructor

Department of Human Sciences

Bogotá, Colombia

*February 2023 - November 2023*

Supervisor: M.Sc Luis Carlos Trujillo

- Taught introductory Python programming to students in the Faculty of Human Sciences.
- Designed a teaching strategy tailored to non-engineering students.
- Created educational presentations to improve knowledge transfer.
- Introduced basic Git concepts for collaborative project management.

## CERTIFICATIONS

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### Introduction to Numerical Analysis

*Graduated November 2020*

Coursera

[View Certificate](#)

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### Process Engineering for the Energy Transition

*Graduated July 2023*

Universidad nacional de Colombia

[View Certificate](#)

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### MCP: Build Rich-Context AI Apps

*October 2025*

DeepLearning.AI

[View Certificate](#)

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### Sustainable Supply Chain Management

*Graduated December 2025*

MITx

[View Certificate](#)

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### Machine Learning with Python: from Linear Models to Deep Learning

*Graduated December 2025*

MITx

[View Certificate](#)

## PUBLICATIONS

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Juan Miguel Mantilla González Julián Esteban Olejua Pinto, *Analysis of Methane Sources and Sinks in Colombia Using Copernicus S5P Data*, En publicación, 2025 [\[Link\]](#).